

9.4 Singular Value Decomposition

In this section we discuss an extension of the diagonalization theory for $n \times n$ symmetric matrices to general $m \times n$ matrices. The results have applications to compression, storage, and transmission of digitized information and form the basis for many of the best computational algorithms currently available for solving linear systems.

Decompositions of Square Matrices

Recall from Section 7.2 that every symmetric matrix A with real entries has an **eigenvalue decomposition**

$$A = PDP^T \quad (1)$$

where P is orthogonal (columns are eigenvectors of A) and D is diagonal (eigenvalues of A). For non-symmetric $n \times n$ matrices, two further decompositions exist:

- **Hessenberg decomposition** (Theorem 7.2.4): $A = PHP^T$ (P orthogonal, H upper Hessenberg).
- **Schur decomposition** (Theorem 7.2.3, when A has real eigenvalues): $A = PSP^T$ (P orthogonal, S upper triangular).

These are important numerically because orthogonal matrices do not magnify roundoff error ($\|P\mathbf{x} - P\hat{\mathbf{x}}\| = \|\mathbf{x} - \hat{\mathbf{x}}\|$).

For a general $m \times n$ matrix A , the factorization of the form

$$A = U\Sigma V^T$$

where U and V are orthogonal (but generally different) is called the **singular value decomposition** (SVD) of A .

Singular Values

Since matrix products of the form $A^T A$ play a central role, we start with two theorems.

Theorem 9.4.1

If A is an $m \times n$ matrix, then:

- (a) A and $A^T A$ have the same null space.
- (b) A and $A^T A$ have the same row space.
- (c) A^T and $A^T A$ have the same column space.
- (d) A and $A^T A$ have the same rank.

Proof of (a). We show every solution of $A\mathbf{x} = \mathbf{0}$ is also a solution of $A^T A\mathbf{x} = \mathbf{0}$, and vice versa. If $A\mathbf{x}_0 = \mathbf{0}$, then $A^T A\mathbf{x}_0 = A^T(\mathbf{0}) = A^T \mathbf{0} = \mathbf{0}$. Conversely, if $A^T A\mathbf{x}_0 = \mathbf{0}$, then $\mathbf{x}_0$ is orthogonal to every vector in $\text{col}(A^T A) = \text{col}(A^T)$ (the row space of A), so $\mathbf{x}_0$ is in the null space of A . Alternatively:

$$\mathbf{x}_0^T (A^T A)\mathbf{x}_0 = (A\mathbf{x}_0)^T (A\mathbf{x}_0) = \|A\mathbf{x}_0\|^2 = 0 \implies A\mathbf{x}_0 = \mathbf{0}. \quad \blacksquare$$

Theorem 9.4.2

If A is an $m \times n$ matrix, then:

- (a) $A^T A$ is orthogonally diagonalizable.
- (b) The eigenvalues of $A^T A$ are nonnegative real numbers.

Proof. (a) $A^T A$ is symmetric, so Theorem 7.2.1 applies. (b) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthonormal eigenbasis for $A^T A$ with eigenvalues $\lambda_1, \dots, \lambda_n$. For $1 \leq i \leq n$:

$$\|A\mathbf{v}_i\|^2 = A\mathbf{v}_i \cdot A\mathbf{v}_i = \mathbf{v}_i \cdot A^T A\mathbf{v}_i = \mathbf{v}_i \cdot \lambda_i \mathbf{v}_i = \lambda_i \|\mathbf{v}_i\|^2 = \lambda_i \geq 0. \quad \blacksquare$$

We assume throughout that the eigenvalues of $A^T A$ are ordered so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$.

Definition 1

If A is an $m \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of $A^T A$ (in nonincreasing order), then the numbers

$$\sigma_1 = \sqrt{\lambda_1}, \quad \sigma_2 = \sqrt{\lambda_2}, \quad \dots, \quad \sigma_n = \sqrt{\lambda_n}$$

are called the **singular values** of A .

EXAMPLE 1 | Singular Values

Find the singular values of $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$.

SOLUTION:

$$A^T A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Characteristic polynomial: $\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0$, so $\lambda_1 = 3$, $\lambda_2 = 1$. The singular values of A are

$$\sigma_1 = \sqrt{3}, \quad \sigma_2 = 1.$$

Singular Value Decomposition

The **main diagonal** of an $m \times n$ matrix is the sequence of entries $a_{11}, a_{22}, \dots$ as far as the matrix extends diagonally (see Figure 9.4.1). Entries on the main diagonal are called **diagonal entries**.

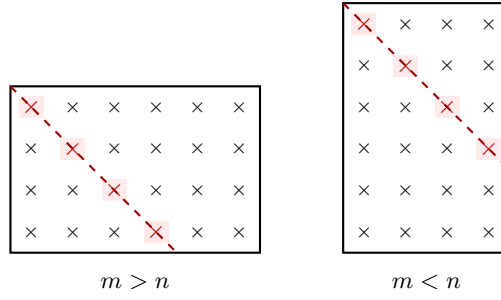


FIGURE 9.4.1 Main diagonals of non-square matrices

Theorem 9.4.3 — Singular Value Decomposition (Brief Form)

If A is an $m \times n$ matrix of rank k , then A can be expressed as $A = U\Sigma V^T$, where Σ has size $m \times n$ and partitioned form

$$\Sigma = \left[\begin{array}{c|c} D & \mathbf{0}_{k \times (n-k)} \\ \hline \mathbf{0}_{(m-k) \times k} & \mathbf{0}_{(m-k) \times (n-k)} \end{array} \right]$$

$D = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_k)$ contains the nonzero singular values in nonincreasing order, U is an $m \times m$ orthogonal matrix, and V is an $n \times n$ orthogonal matrix.

Theorem 9.4.4 — Singular Value Decomposition (Expanded Form)

If A is an $m \times n$ matrix of rank k , then A can be factored as

$$A = U\Sigma V^T = [\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_k \mid \mathbf{u}_{k+1} \ \cdots \ \mathbf{u}_m] \begin{bmatrix} \sigma_1 & & & & \mathbf{0} \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_k & \\ \hline & \mathbf{0} & & & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \\ \mathbf{v}_{k+1}^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix}$$

in which U , Σ , V have sizes $m \times m$, $m \times n$, $n \times n$, and:

- (a) $V = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n]$ orthogonally diagonalizes $A^T A$.
- (b) Nonzero diagonal entries of Σ are $\sigma_i = \sqrt{\lambda_i}$, where $\lambda_1, \dots, \lambda_k$ are the nonzero eigenvalues of $A^T A$ corresponding to column vectors of V .
- (c) Columns of V are ordered so $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0$.
- (d) $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|} = \frac{1}{\sigma_i} A\mathbf{v}_i \quad (i = 1, 2, \dots, k)$.
- (e) $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is an orthonormal basis for $\text{col}(A)$.
- (f) $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_m\}$ is an orthonormal basis for R^m .

The vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are **left singular vectors** and $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **right singular vectors** of A .

Proof. We prove the theorem for the case where A is $n \times n$; the rectangular case requires only notational adjustments.

Step 1: Orthonormal eigenbasis for $A^T A$. By Theorem 9.4.2(a), $A^T A$ is orthogonally diagonalizable. Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthonormal eigenbasis for R^n with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$. Since $\text{rank}(A) = k$, Theorem 9.4.1(d) gives $\text{rank}(A^T A) = k$, so exactly k eigenvalues are positive: $\lambda_1 \geq \cdots \geq \lambda_k > 0$ and $\lambda_{k+1} = \cdots = \lambda_n = 0$.

Step 2: $\{A\mathbf{v}_1, \dots, A\mathbf{v}_k\}$ is an orthogonal set of nonzero vectors. For $1 \leq i \leq k$:

$$\|A\mathbf{v}_i\|^2 = \mathbf{v}_i \cdot A^T A \mathbf{v}_i = \mathbf{v}_i \cdot \lambda_i \mathbf{v}_i = \lambda_i \|\mathbf{v}_i\|^2 = \lambda_i > 0$$

so each $A\mathbf{v}_i \neq \mathbf{0}$. For $i \neq j$ (both $\leq k$):

$$A\mathbf{v}_i \cdot A\mathbf{v}_j = \mathbf{v}_i \cdot A^T A \mathbf{v}_j = \mathbf{v}_i \cdot \lambda_j \mathbf{v}_j = \lambda_j (\mathbf{v}_i \cdot \mathbf{v}_j) = 0$$

since $\{\mathbf{v}_i\}$ is orthonormal. For $i > k$: $\|A\mathbf{v}_i\|^2 = \lambda_i = 0$, so $A\mathbf{v}_i = \mathbf{0}$.

Step 3: $S = \{A\mathbf{v}_1, \dots, A\mathbf{v}_k\}$ is an orthogonal basis for $\text{col}(A)$. S consists of k nonzero orthogonal vectors, hence linearly independent. Since each $A\mathbf{v}_i \in \text{col}(A)$ and $\dim \text{col}(A) = k$, S is a basis for $\text{col}(A)$.

Step 4: Left singular vectors (part (d)) and orthonormal basis for $\text{col}(A)$ (part (e)). Define

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|} = \frac{1}{\sqrt{\lambda_i}} A\mathbf{v}_i = \frac{1}{\sigma_i} A\mathbf{v}_i \quad (i = 1, 2, \dots, k)$$

Normalizing the orthogonal basis S gives the orthonormal set $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$, which is therefore an orthonormal basis for $\text{col}(A)$. Equivalently,

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i \quad (i = 1, \dots, k) \quad (4)$$

Step 5: Orthonormal basis for R^m (part (f)). By the Gram–Schmidt process (Theorem 6.3.6), extend $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ to an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_m\}$ for R^m . Let $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ (orthogonal, $m \times m$) and $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ (orthogonal, $n \times n$).

Step 6: $A = U\Sigma V^T$. It suffices to show $AV = U\Sigma$, since $V^{-1} = V^T$ then gives $A = U\Sigma V^T$. Compare column i of each side:

- For $i \leq k$: column i of AV is $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ (by (4)), and column i of $U\Sigma$ is $\sigma_i \mathbf{u}_i$ (the i th column of Σ has σ_i in row i and zeros elsewhere, so $U\Sigma \mathbf{e}_i = \sigma_i \mathbf{u}_i$). ✓
- For $i > k$: column i of AV is $A\mathbf{v}_i = \mathbf{0}$ (Step 2), and column i of $U\Sigma$ is $\mathbf{0}$ (the i th column of Σ is all zeros). ✓

Hence $AV = U\Sigma$, and multiplying on the right by V^T gives $A = U\Sigma V^T$. ■

EXAMPLE 2 | Singular Value Decomposition (Non-Square A)

Find a singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$.

SOLUTION: From Example 1, $\lambda_1 = 3$, $\lambda_2 = 1$ and $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$. One can verify that

$$\mathbf{v}_1 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}$$

are eigenvectors of $A^T A$ corresponding to $\lambda_1 = 3$ and $\lambda_2 = 1$, so $V = [\mathbf{v}_1 \mid \mathbf{v}_2]$ orthogonally diagonalizes $A^T A$.

From Theorem 9.4.4(d):

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} A\mathbf{v}_1 = \frac{\sqrt{3}}{3} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{1} A\mathbf{v}_2 = \begin{bmatrix} 0 \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

To find $\mathbf{u}_3$, we need a unit vector orthogonal to both $\sqrt{6}\mathbf{u}_1 = (2, 1, 1)^T$ and $\sqrt{2}\mathbf{u}_2 = (0, -1, 1)^T$. Solving the homogeneous system gives general solution $t(-1, 1, 1)^T$, so normalizing:

$$\mathbf{u}_3 = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

Thus the SVD of A is

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\sqrt{6}}{3} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{1}{\sqrt{3}} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{1}{\sqrt{3}} \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}}_{V^T}$$

Exercise Set 9.4

In Exercises 1–4, find the distinct singular values of A .

1. $A = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$

2. $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

4. $A = \begin{bmatrix} \sqrt{2} & 0 \\ 1 & \sqrt{2} \end{bmatrix}$

In Exercises 5–8, find a singular value decomposition of A .

5. $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

6. $A = \begin{bmatrix} -3 & 0 \\ 0 & -4 \end{bmatrix}$

7. $A = \begin{bmatrix} 4 & 6 \\ 0 & 4 \end{bmatrix}$

9. $A = \begin{bmatrix} -2 & 2 \\ -1 & 1 \\ 2 & -2 \end{bmatrix}$

Working with Proofs

13. Prove: If A is an $m \times n$ matrix, then A and $A^T A$ have the same rank.
14. Prove part (d) of Theorem 9.4.1 by using part (a) and the fact that A and $A^T A$ have n columns.
17. Prove that the singular values of $A^T A$ are the squares of the singular values of A .
18. Prove: If $A = U \Sigma V^T$ is a singular value decomposition of A , then U orthogonally diagonalizes AA^T .

19. A **polar decomposition** of an $n \times n$ matrix A is a factorization $A = PQ$ where P is positive semidefinite with the same rank as A , and Q is orthogonal.
- a. Prove: If $A = U\Sigma V^T$ is a singular value decomposition of A , then $A = (U\Sigma U^T)(UV^T)$ is a polar decomposition of A .
- b. Find a polar decomposition of the matrix in Exercise 5.